

# Mikel Val Calvo PhD

Research Engineer | Applied AI, RAG and Agentic Systems

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Crevillente, Spain

## PROFILE

AI consultant and applied researcher with a PhD, an MSc in Advanced AI, and experience spanning LLM-based agents, retrieval-augmented generation (RAG), knowledge graphs and database integration. Previously contributed to data-agent design for Inditex and led MindFrameAI at LabLENI. Combines Python engineering with experimental design, multidisciplinary research and peer-reviewed publications, including Ontogenix for LLM-driven ontology engineering.

## FEATURED RESEARCH

**Ontogenix:** LLM-based pipeline for ontology engineering from datasets (2023).

[github.com/tecnomod-um/OntoGenix](https://github.com/tecnomod-um/OntoGenix)

Val-Calvo, M., et al. (2025). Ontogenix: Leveraging large language models for enhanced ontology engineering from datasets. Information Processing & Management, 62(3), 104042.

## TECHNICAL SKILLS

**LLMs and agents:** Agentic AI, RAG, ReAct, LangGraph, LangChain, LLM orchestration, prompt engineering, self-repairing mechanisms, OpenAI, Hugging Face, NLP.

**Evaluation and observability:** OpenTelemetry, Arize Phoenix.

**Data and ML:** Python (advanced, OOP), PyTorch, Transformers, Pandas, NumPy, SciPy, Scikit-learn, knowledge graphs, ontologies, database integration.

**Engineering:** GitHub, Azure DevOps, CI/CD, Docker, real-time systems.

## RELEVANT EXPERIENCE

### AI Consultant

2025 - Present

TMC People Drive Technology | Previous assignment: Inditex AI Team

- Previously contributed to the design of an LLM-based data agent that orchestrates business data and enables natural-language queries for executives, commercial teams and marketing stakeholders.

### Senior Researcher

2024 - 2025

LabLENI, HUMAN-TECH Research Center | Universidad Politécnica de Valencia

- Led MindFrameAI research on cognitive architectures, empathetic agents and multi-agent systems; coordinated technical work across multidisciplinary projects.
- Conceived generative AI for managing interviews and analysing results in ASISVIA.
- Contributed to concepts and high-level software architecture for agent-based systems and immersive training.

Ongoing research collaboration: empathetic agents and AI civilizations with LabLENI and the National University of Singapore, exploring emergent social behavior in multi-agent systems.

### Postdoctoral Researcher

2023 - 2024

Universidad de Murcia | BASF-funded research

- Developed LLM-based systems with OpenAI and LangChain to translate structured databases into ontologies and knowledge graphs for data interoperability and access.
- Designed graph-based database integration using multi-agent methods, prompting, RAG and self-repairing mechanisms; managed project work with cross-functional BASF teams on implementation and deployment.

## EARLIER RESEARCH AND ENGINEERING

### Postdoctoral Researcher

2020 - 2022

Neuroprosthesis and Visual Rehabilitation Unit | Universidad Miguel Hernández

- Led a team developing and validating a cortical visual neuroprosthetic device. Designed experimental protocols and custom software, and built a wearable prototype for image processing and cortical electrical stimulation.
- Applied machine learning, deep learning and computer vision to neuroprosthetics, combining experimental validation with complex prototype development.

### Doctoral Researcher

2017 - 2020

Universidad Politécnica de Cartagena

- Designed and implemented a multimodal emotion-recognition system for human-robot interaction, combining deep-learning ensembles for facial expressions with physiological signal analysis; presented results at international conferences.

**Earlier roles:** Assistant Researcher, Universidad Politécnica de Cartagena (2016 - 2017), integrating biological neural networks with robot control and computer vision; Analyst Programmer, Universidad Miguel Hernández (2015 - 2016), developing ROS-based vision algorithms for obstacle-free robotic navigation.

## RESEARCH SOFTWARE AND PUBLICATIONS

**Experimental research software:** Biosignals, GEERT and GePHYCAM for real-time physiological, EEG and camera-based experiments; NeuroSorter for CNN-based spike/noise discrimination and clustering.

Selected earlier publications: real-time multimodal emotion estimation (International Journal of Neural Systems, 2020); affective human-robot interaction (IEEE Access, 2020); real-time EEG artifact removal and emotion estimation (Frontiers in Computational Neuroscience, 2019).

Publication record: [researchgate.net/profile/Mikel\\_Calvo2](https://researchgate.net/profile/Mikel_Calvo2)

## TEAM AWARDS

**2025 | UPV Excellence Award for Knowledge Transfer.** Member of the NeuroCash team recognised for banknote neurodesign in collaboration with Banco de España.

**2022 | Onda Cero Science and Research Award.** Member of the UMH Biomedical Neuroengineering Group recognised for developing a cortical visual implant based on microelectrodes.

## EDUCATION

**PhD | UNED - UPCT - UMH | 2017 - 2020**

Thesis: Emotional Human-Robot Interaction Using Physiological Signals.

**MSc in Advanced Artificial Intelligence | UNED | 2016 - 2017**

**BSc in Software Engineering | UNED | 2010 - 2015**

## PROFESSIONAL DEVELOPMENT

2026 | A2A: The Agent2Agent Protocol - Google Cloud & IBM Research; Fine-tuning & RL for LLMs: Intro to Post-training - DeepLearning.AI.

2023 | ChatGPT Prompt Engineering for Developers - DeepLearning.AI / Coursera.

2018 | DeepLearning.AI / Coursera courses: Neural Networks and Deep Learning; Convolutional Neural Networks; Hyperparameter Tuning, Regularization and Optimization; Structuring Machine Learning Projects; Sequence Models.

## LANGUAGES AND SCIENTIFIC COMMUNITY

English, Spanish and Basque (fluent/native); Valencian (conversational). Co-organizer of the IWINAC/ICINAC Congress on natural and artificial computation.